

Zaighum Zawar

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with a strong foundation in Machine Learning, modern AI technologies and Full-Stack Web Development. Experienced in designing, developing, and deploying AI-powered applications, Building Generative AI, Agentic AI systems, and Retrieval-Augmented Generation (RAG) pipelines. Proficient in Python, JavaScript, Langchain, LangGraph React, Node.js, FastAPI, MCPs and RESTful APIs, with hands-on experience in Context Engineering, prompt engineering, vector databases, and exposure to cloud platforms (AWS/GCP/Azure). Skilled in building scalable, high-performance, and user-centric solutions, with a focus on MLOps, Agentic AI Solutions, system design, and real-world problem solving

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn), JavaScript, SQL, PHP, HTML/CSS
AI / ML: LangChain, LangGraph, RAG, Vector Databases (ChromaDB), Prompt Engineering, Context Engineering, FastMCP
Web & APIs: FastAPI, React, Node.js, RESTful APIs, WebSockets, Microservices, Docker
Databases & Cloud: PostgreSQL, MySQL, MongoDB, GCP, AWS, Azure, Git/GitHub
Tools: N8N, Antigravity, Claude Code, Cursor, Jupyter Notebook, Postman, Linux, GitHub Copilot, MCPs, VS Code

PROFESSIONAL EXPERIENCE

Co-Founder & AI Engineer Jan 2024 – Present
Botanex Forge Remote

- Led development of advanced automation workflows and Python automation bots that streamlined client business operations end-to-end.
- Co-developed and deployed full-stack web and mobile applications for clients using React, FastAPI, and PostgreSQL on cloud infrastructure.
- Designed and integrated Agentic AI systems and LLM-powered pipelines into client products leveraging LangChain, LangGraph, and MCP protocols.

PROJECTS

Agentic Clinical Decision Support System (CDSS) *Python, LangChain, LangGraph, RAG, FastMCP, ChromaDB*

- Built an AI-driven CDSS using agentic reasoning loops and RAG pipelines to provide clinicians with evidence-based, real-time decision support.
- Integrated vector database retrieval (ChromaDB) over medical knowledge corpora, enabling accurate context-grounded clinical recommendations.

Blog Generation Agentic AI App *FastAPI, LangGraph, LangChain, Groq*

- Engineered an Agentic AI blog generator using FastAPI and LangGraph with graph-based routing for topic-driven automated content generation.
- Implemented multilingual translation paths (French and Urdu) via LLM chaining, expanding content accessibility across diverse audiences.

AI-Powered Vehicle Predictive Health Maintenance *Python, WebSockets, PostgreSQL, pgvector, RAG*

- Built a real-time vehicle diagnostics platform ingesting live OBD-II data via WebSockets into an agentic AI reasoning core.
- Implemented RAG with pgvector for car-specific fault analysis and repair recommendations, improving diagnostic accuracy and speed.

EDUCATION

National University of Computer and Emerging Sciences (NUCES FAST) Aug 2022 – Present
Bachelor of Science in Computer Science (BsCS)
The City School, Gulshan-e-Iqbal Campus Jun 2020 – Jun 2022
A Levels (Advanced Levels)
The City School, Johar Campus Jun 2017 – Jun 2020
O Levels (Ordinary Levels)

PROFESSIONAL CERTIFICATIONS

GCP Generative AI Leader – Google Cloud Platform
IBM Agentic AI with LangChain and LangGraph – IBM
IBM AI Agents and Agentic Workflows – IBM
Machine Learning Specialization – Coursera / DeepLearning.AI
GitHub Foundations Certification – GitHub
Google Prompting Essentials – Google
MongoDB Associate Developer – MongoDB
Web Designing for Everyone – University of Michigan